# ■ Supplementary document

Hello. Without light, you see nothing. But where does the light come from and how does it reach your eyes? The moon shines in the sky, but it does not produce any light. She just pretends to shine. In this video, you will see how you see an object, how you travel the light, how to predict a shadow and how to protect your eyes. Part one, see an object. A primary source produces the light it has. The sun, a lamp lit, the flame of a candle, a screen lit. The Moon does not produce light. She receives the sun and sends it back in all directions. They say she's broadcasting it. It's a diffuse object. How do you see a book? A source enlightens him, the book spreads this light and a part enters your eye. Without light coming into your eye, you see nothing. Part two, the path of light. In the air, light spreads in a straight line. It is represented by a ray of light, a straight line with an arrow. You see it when the sun crosses the mist, the rays are straight. A beam of light is a set of rays from the same source. It is represented by these two limit rays. The light goes very fast, about 300,000 km per second. It takes a little more than 8 minutes to come from the sun and about a second, a third from the moon. Part three, provide a shadow. The light does not pass through an opaque object. Behind him, there is an area without light, it is the shadow. Place a small lamp near an object. The rays around its edges spread a lot and the shadow on the screen is large. Get the lamp away. The boundary rays become almost parallel and the shadow narrows to near the size of the object. The light beam model thus predicts the size of the number with a simple diagram. Part four, the dangers of light. Overly intense light can cause permanent damage to the eye, either directly or after reflection. Three cases, the sun, a class 3 or 4 laser and the electric arc of a solder. To observe an eclipse, you need special glasses for that. Ordinary sunglasses are not enough. So what's the point? All this? Not just to pass control. An architect predicts the shadow of a building on its neighbours. An optician makes glasses following the path of light rays. An astronomer studies the light coming from the stars. And a padder protects himself from the bow with a mask. These concepts are used in building, vision health, astronomy and industry. To sum up, one sees an object here of light coming from him enters the eye. The light goes straight, very fast. An opaque object is a shadow. And some lights damage the eye forever. Your turn. Why do we see the moon? Put the video on hold. Because it diffuses sunlight. It's a diffuse object, not a primary source. Next time you see a shadow, think about the rays that draw it. Subscribe for the rest and see you soon.